Response to Reviewers

CICT-2026, Paper ID 760
A Unified Ingestion-to-Delivery Pipeline for Converting Multi-View 2D Media into 3D Assets

We thank both reviewers. Each numbered item below is a requested revision, with what was done and where the evidence is. Every number cited traces to a result file in the repository; labels such as B4 or B10 name the experiment behind it.

Reviewer 1

1. Roadmap for the missing real-time delivery protocols (RTSP/WebRTC). This is a roadmap, so it lives here and not in the paper, which keeps its limitation statement. The path from where the prototype is:

1. *RTSP ingestion.* The live track already takes WebSocket frames into a bounded ring buffer with backpressure. RTSP is another producer into the same buffer: an FFmpeg-backed reader that decodes a stream and pushes frame references, so the buffer, priority tagging and schema are unchanged.
2. *WebRTC ingestion.* A signalling endpoint on the existing gateway (reusing its authentication and rate limiting) and a receiver that terminates the peer connection and feeds the same buffer.
3. *Live reconstruction.* The current reconstruction stage needs at least two views and an offline alignment, so a live path needs an incremental one: a sliding window over the buffer, aligned with the existing windowed pairing and re-fused as frames arrive. This is the largest piece and is future work.
4. *Live delivery.* The live outcome route, which the routing contract rejects today, is added once step 3 produces geometry; the contract already isolates it, so the other routes are unaffected.

2. Future work should prioritise quantitative comparison with classical MVS and MASt3R baselines. Kept as the first priority in the Limitations section. The paper now reports ground-truth accuracy for the pipeline itself (see Reviewer 2, item 1), and says plainly that no baseline ran under the same protocol.

Reviewer 2

1. Ground-truth accuracy (Chamfer, F-score, normal consistency) on DTU and CO3D; baselines; the effect of retention order.

- *DTU.* Five scans (scan24, 37, 40, 55, 63, labelled S1–S5 in that order in the paper's tables), DTU's own evaluation protocol, with a similarity fit to the calibrated cameras that is trimmed of lost views (B2; table in the paper). The effect of retention order is B4: restoring capture order did not improve reconstruction.
- *Normal consistency on DTU (new).* DTU's ground truth has no normals, so they are estimated from the structured-light points and compared with the reconstructed meshes' normals (B10). The measured value is about 0.52 mean $|\cos|$ for every variant, close to the 0.5 of unrelated normals, so it does not separate the variants. We therefore report Chamfer, F-score and DTU accuracy/completeness in the paper and leave this metric in the repository, not as a result.
- *CO3D (new).* CO3D's point cloud is a COLMAP reconstruction, not an independent ground truth, and the single-sequence release has no camera annotations. We therefore score agreement with CO3D's reference after registering to it (a fixed scale, then ICP), report the registration residual (1.4–1.5% of the bounding-box diagonal), and call it a pseudo-reference (B11). Refined and isolated, the mesh scores F@2% 0.745 at 8 views and 0.769 at 24, against 0.586 and 0.502 for the raw point cloud (new subsection "Further Evaluation" in the paper).

- *Baselines.* Not done in this revision. We report that no baseline ran under the same protocol and do not compare against published MVS scores, which are given DTU's cameras. The second backend below (DUSt3R) is scored under the same protocol, so one comparison now exists: at ten views MASt3R reaches mean F-score 0.210 and 5.85 mm overall error, DUSt3R 0.069 and 8.3 mm.

2. Validate frame selection and synchronization on real multi-camera data. We used five real DTU scans (scan24, 37, 40, 55, 63), each a real 49-view capture from a camera moved between positions by a robot arm. Frame selection and its effect on reconstruction are B4 on those captures. For synchronization (B12), the images and the motion signal are real: the production sampler's motion score is computed on consecutive real frames, and a second camera's scores come from its own perturbed pixels (gain, offset, blur, sensor noise). What is injected is the clocks, since DTU's cameras have no timestamps; recovery is scored against the injected offset. Over 1,100 trials the median absolute error is 12–30 ms per scan against a 40 ms frame interval, the sign is right in 84–97% of nonzero-offset trials and groups form in 83–90%; a minority of trials fail by much more than the median, so the mean error (31–145 ms) is larger. This is a validation of the synchronizer on a real scene's signal, not a phone-rig experiment, and the paper says so. We did not collect new multi-camera recordings or test mixed-media payloads, which the ingestion router rejects by design.

3. Batched component filter, rerun on several real meshes with variance, and justification for dropping 78% of cells.

- *Batched filter.* Implemented on the GPU and checked against the original on the synthetic ladder and the CO3D trace.
- *Real meshes with variance (new, B10).* Both filters timed on the raw meshes of scan24, 37, 40, 55 and 63 (1.3–2.6M cells each) with five repeats. The batched filter takes 0.30–0.84 s (SD at most 0.09 s). The per-body filter finished only on scan37 (1,259 s, with identical output); on the other four it exceeded the 30-minute budget. Refinement is deterministic, so the variance is timing only.
- *The dropped cells (new, B10).* On DTU, object mode keeps 11–26% of the faces and on average scores lower than the raw mesh, so we do not justify the drop as an accuracy gain. With object isolation on, the dominant support plane is cut first and the subject's fragments gathered; B10 reports what that keeps and how it scores, next to the plain largest-component rule. Isolation cuts the support plane (32–60% of faces) and keeps 15–43% of the faces on four scans, where its mean F-score is 0.239 against 0.229 raw and 0.217 for object mode. On scan37, a pose failure throughout, the score collapses (F-score 0.002); we report this and have not tuned it away.

4. Demonstrate stage replaceability with a second reconstruction backend. Reconstruction now sits behind a backend interface, selected by the job. DUSt3R, a different network with a different optimiser and no TSDF fusion, runs behind it. B13 runs it on the five DTU scans and records, per run, that the stage wrote the same artifacts and that refinement consumed its mesh unchanged. A contract test states what any backend must produce. Replaceability is shown for the reconstruction stage; we do not claim it for every stage. DUSt3R met the artifact contract and the refinement hand-off on all five scans. Its geometry is worse under our protocol (mean F-score 0.069, all five flagged as pose failures), which shows the swap changes the measurement; it is evidence of replaceability, not of an equally accurate backend.

5. Align the title and abstract with the implemented scope. The title now reads "Multi-View 2D Media". The abstract and introduction state that reconstruction accepts image folders, videos and synchronized captures, and that single images and live streams are rejected and are future work.

6. Release code, configurations, seeds, and hardware/thresholds. The code, experiment scripts and every result CSV are in the repository; each result row records its seed, the library versions, the GPU, and the commit. The thresholds are named constants in the source, including the motion thresholds

and the synchronizer's tolerances. Repository: `https://github.com/AyushK0808/z-shift`. Hardware: GPU runs used an NVIDIA GeForce RTX 5060 Laptop GPU (8 GB) under WSL2 on Windows 11 at 512-pixel input, the determinism test also ran on a Tesla T4, and the CPU benchmarks ran on an Intel Core i7-14650HX laptop; the camera-ready version will carry the same statement.